INCH-POUND
MIL-PRF-62411/3A(AT)
7 July 1997
SUPERSEDING
MIL-H-62411/3(AT)
5 July 1985

PERFORMANCE SPECIFICATION SHEET

HEAT EXCHANGER, NBC SYSTEM
PRECOOLER P/N M62411/3
(ENGINEERING 12336768)

This specification is approved for use by the U.S. Army Tank-automotive and Armaments Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-PRF-62411A(AT) listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

REQUIREMENTS

Design and construction. The precooler shall conform to drawing hierarchy of which Drawing 12336768 is the top drawing.

Weight. The weight of the precooler exchanger shall not exceed 45 pounds (lb) (20.4 kilograms (kg)) (see 4.6.3).

Performance.

Leakage.

External leakage. The pneumatic pressure applied to the hot air passages of the heat exchanger shall be 100 pounds per square inch gage (psig) (689 kilopascals (kPa)) (see 4.6.4.1).

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Internal leakage. The pressure applied to the hot air passages shall be 100 psig (see 4.6.4.2).

Proof pressure. The proof pressure shall be 325 psig (2240.7 kPa)) (see 4.6.5).

Burst pressure. The burst pressure shall be 300 psig (2068 kPa) at 845 degrees Fahrenheit (°F) (452 degrees Celsius (°C)) (see 4.6.6).

Heat transfer. Heat transfer performance shall be not less than that shown in figure 1 (see 4.6.7).

Air pressure loss. The pressure loss shall not exceed that shown in figure 2 (see 4.6.8).

Leakage.

External leakage. A pneumatic pressure of 100 to 103 psig (689 to 710 kPa) shall be applied to the hot air passages.

Internal leakage. A pneumatic pressure of 100 to 130 psig shall be applied to the hot air passages.

Proof pressure. The test pressure shall be 325 to 335 psig (2240.7 to 2309.7 kPa).

Burst pressure. The oven temperature shall be 845°F (452°C). The test pressure shall be 300 to 309 psig (2068 to 2130 kPa).

Heat transfer. To determine conformance to 3.5.4, the heat exchanger shall operate at each combination of the steady state conditions specified in table VI. The measured hot air outlet temperature shall be equal to or greater than that specified in table VI.

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TABLE VI. Heat transfer.

Initial air temperature °F (°C)		Air flow lb/min (kg/min)		Hot air outlet outlet temperature
Hot	Cold	Hot	Cold	°F (°C)
300 ±15 (148.9 ±8.3)	Lab ambient	10 (4.5)	15 (6.8)	173 ±15 (78.3 ±8.4)
			30 (13.6)	276 ±15 (135.6 ±8.4)
			52.2 (23.7)	290 ±15 (143.3 ±8.4)
420 ±20 (215.6 ±11)		16.6 (7.5)	15	271 ±20 (132.8 ±11)
			30	336 ±20 (169.4 ±11)
			52.2	374 ±20 (190 ±11)
800 ±40 (426.7 ±22.2)		20 (90.7)	15	460 ±40 (238 ±22)
			30	600 ±40 (316 ±22)
			52.2	680 ±40 (360 ±22)

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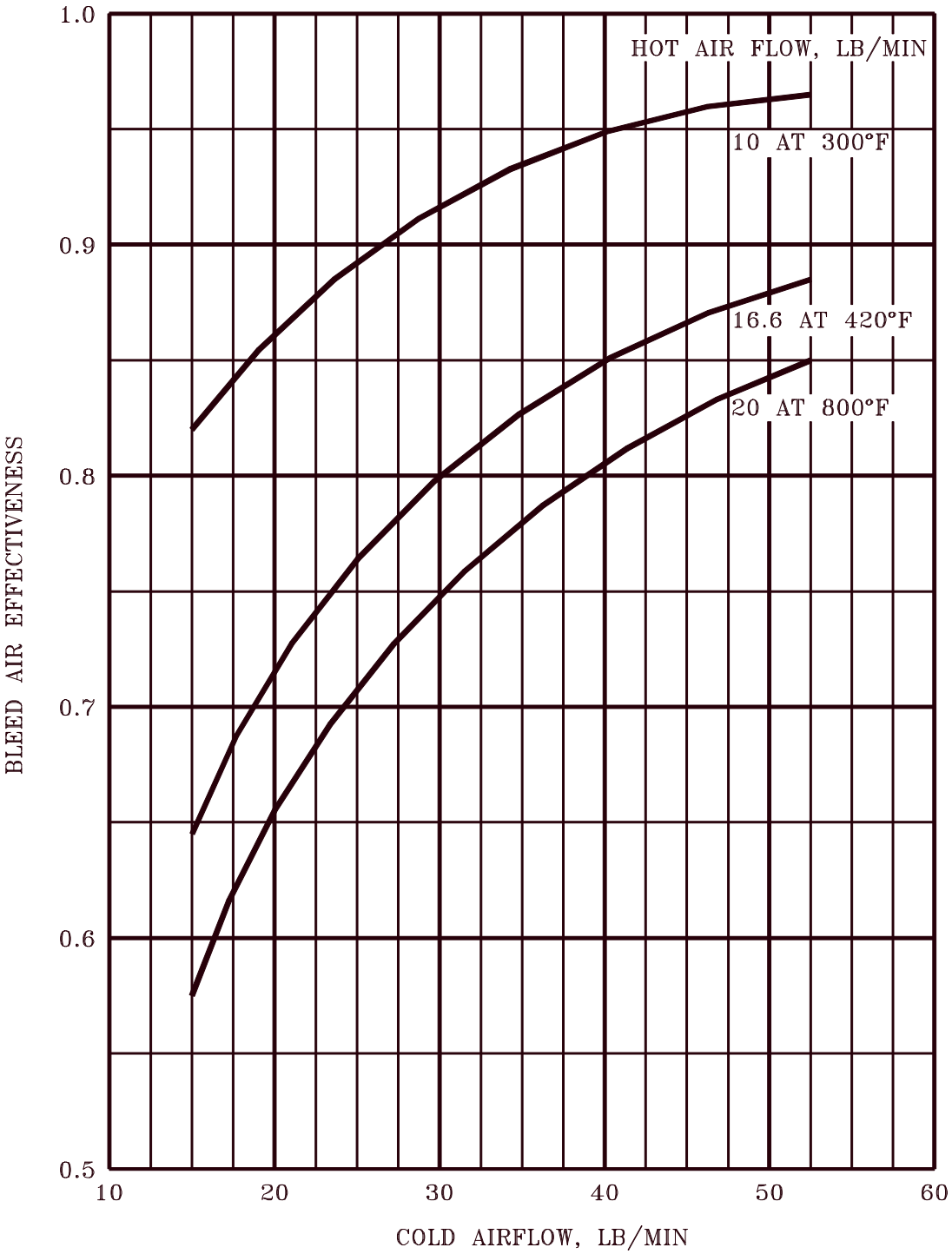


FIGURE 1. Required precooler performance.

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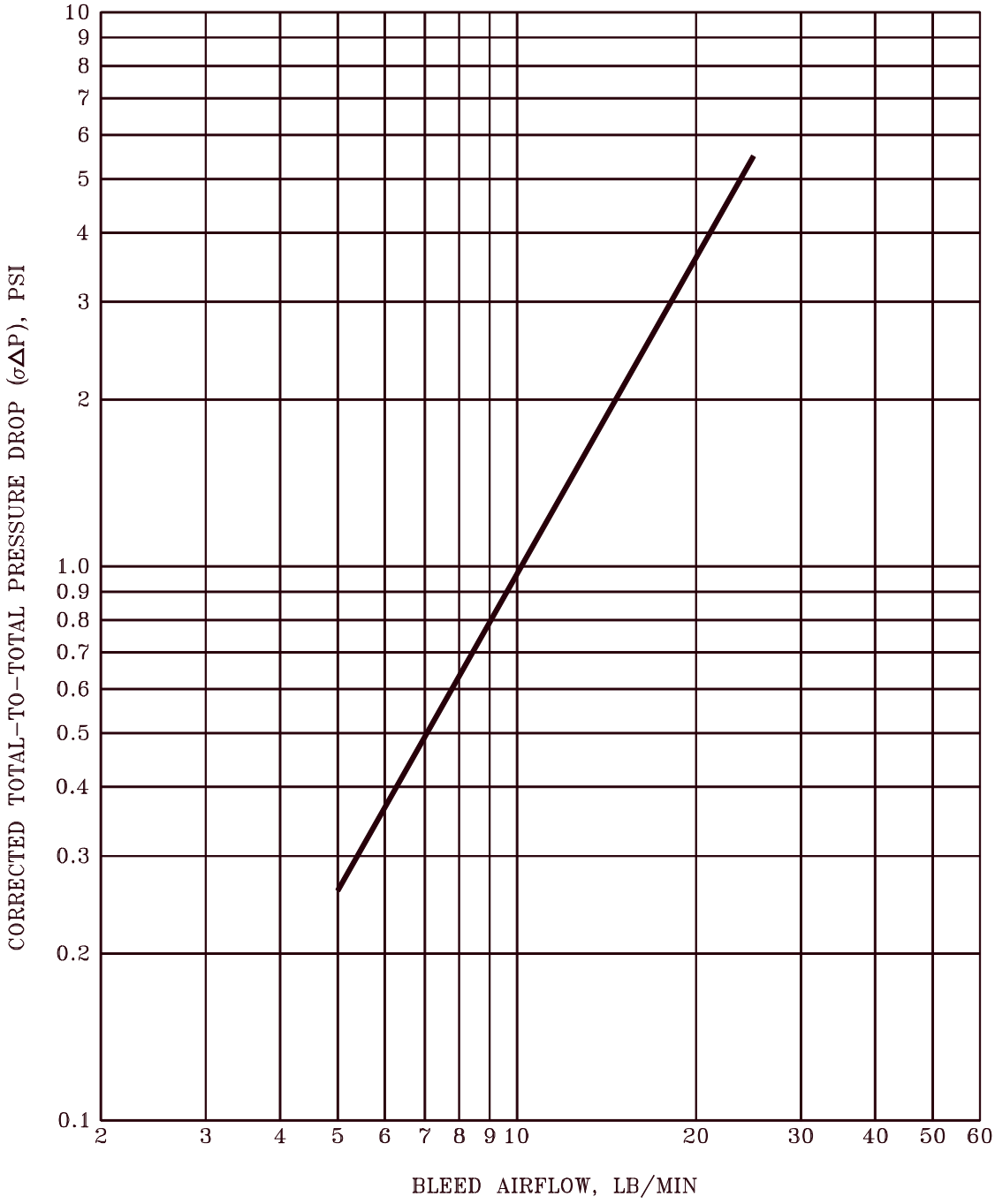
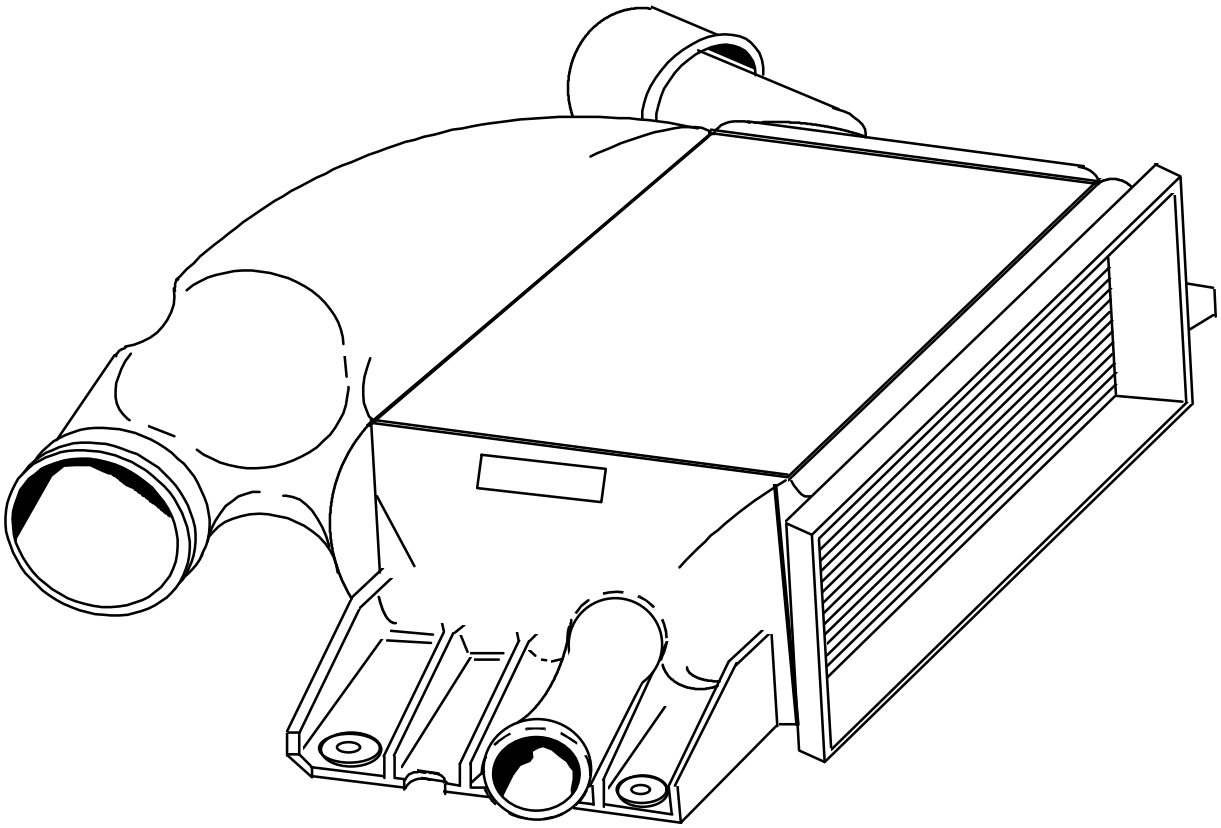


FIGURE 2. Allowable precooler bleed air pressure loss.

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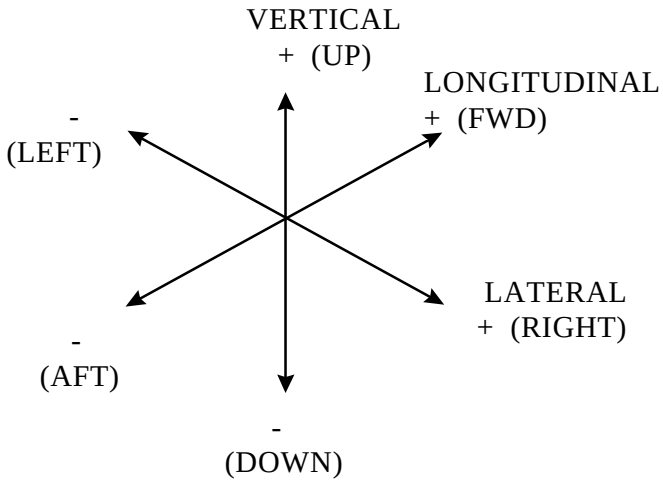


FIGURE 3. Axes orientation.

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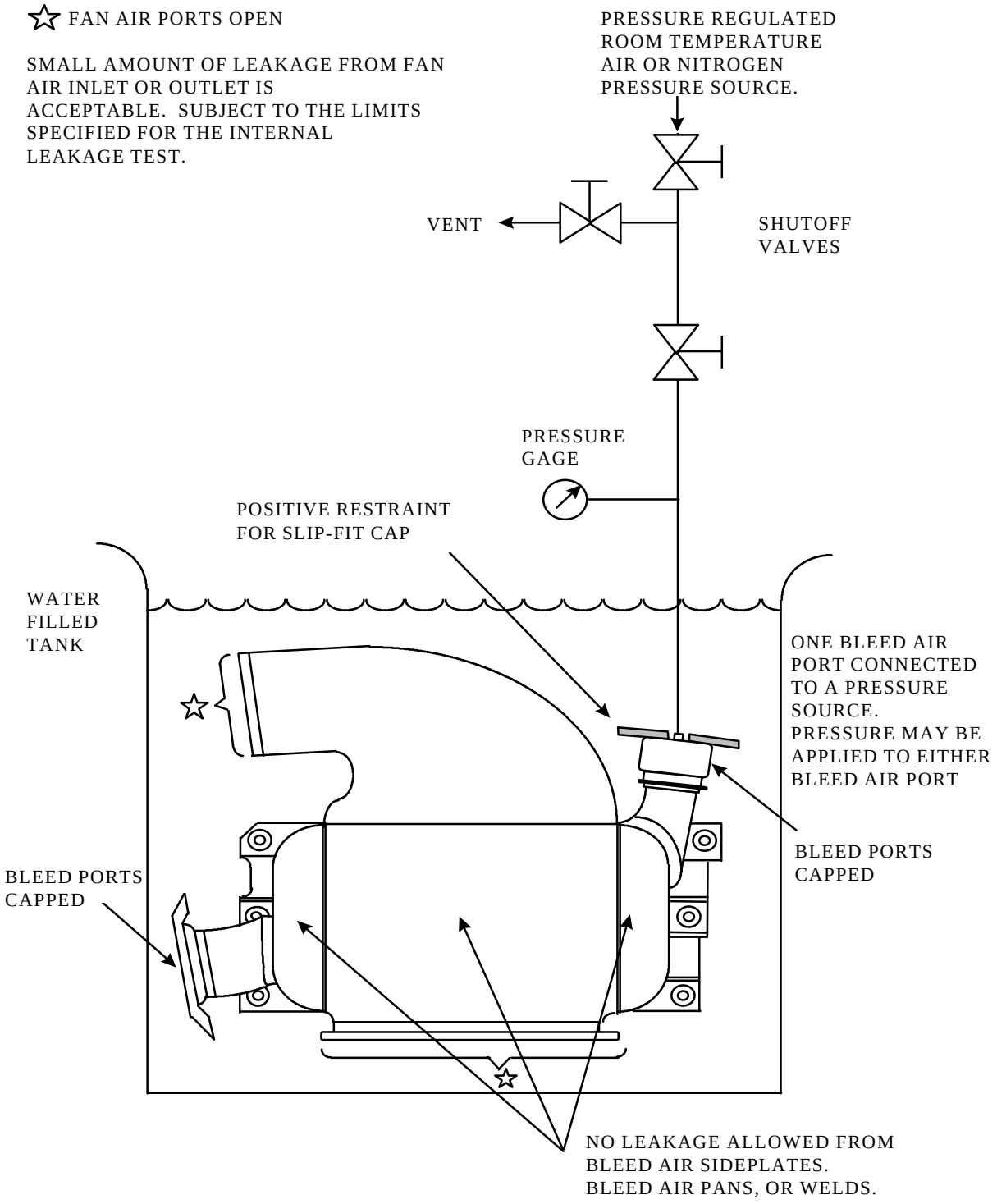


FIGURE 4. Leakage test setup.

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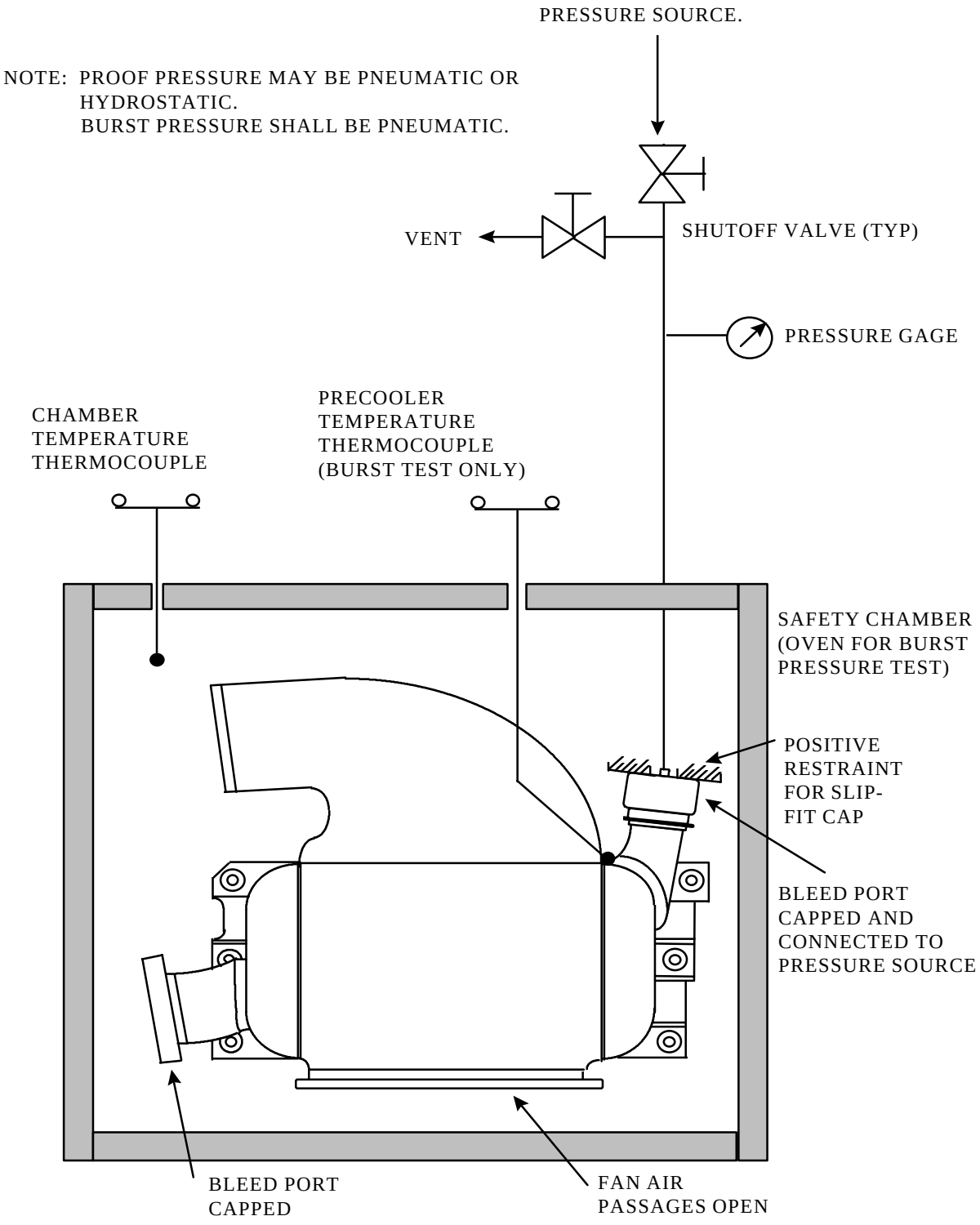


FIGURE 5. Proof and burst pressure test setup.

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